🌐 Module 05: Web Application Penetration Testing

```
# Exercise 2: Performing Website Footprinting Using AI
```

Scenario

In this lab, we will use AI tools such as ChatGPT to generate a Python script for performing comprehensive website footprinting and enumeration. The script will automate tasks such as analyzing HTML source code, identifying server-side technologies, crawling the website, extracting metadata, and testing for protections like WAFs and load balancers.

Objective

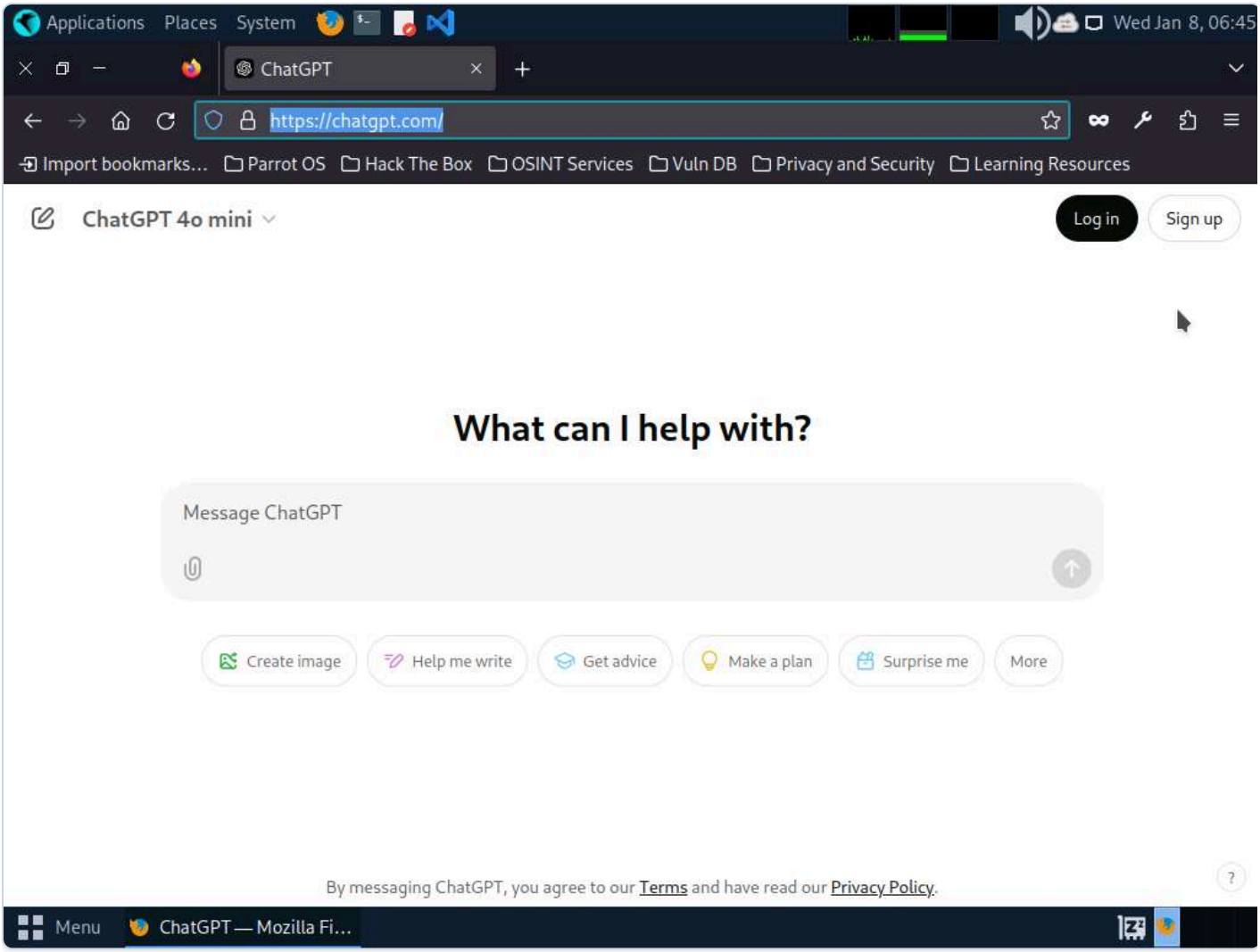
The objective of this lab is to help students learn how to leverage AI models to generate customized Python scripts for comprehensive website footprinting and security analysis.

Task 1: Performing Website Footprinting Using AI

In this task, we generate and execute a custom Python script to perform automated website footprinting on the target domain `www.luxurytreats.com`.

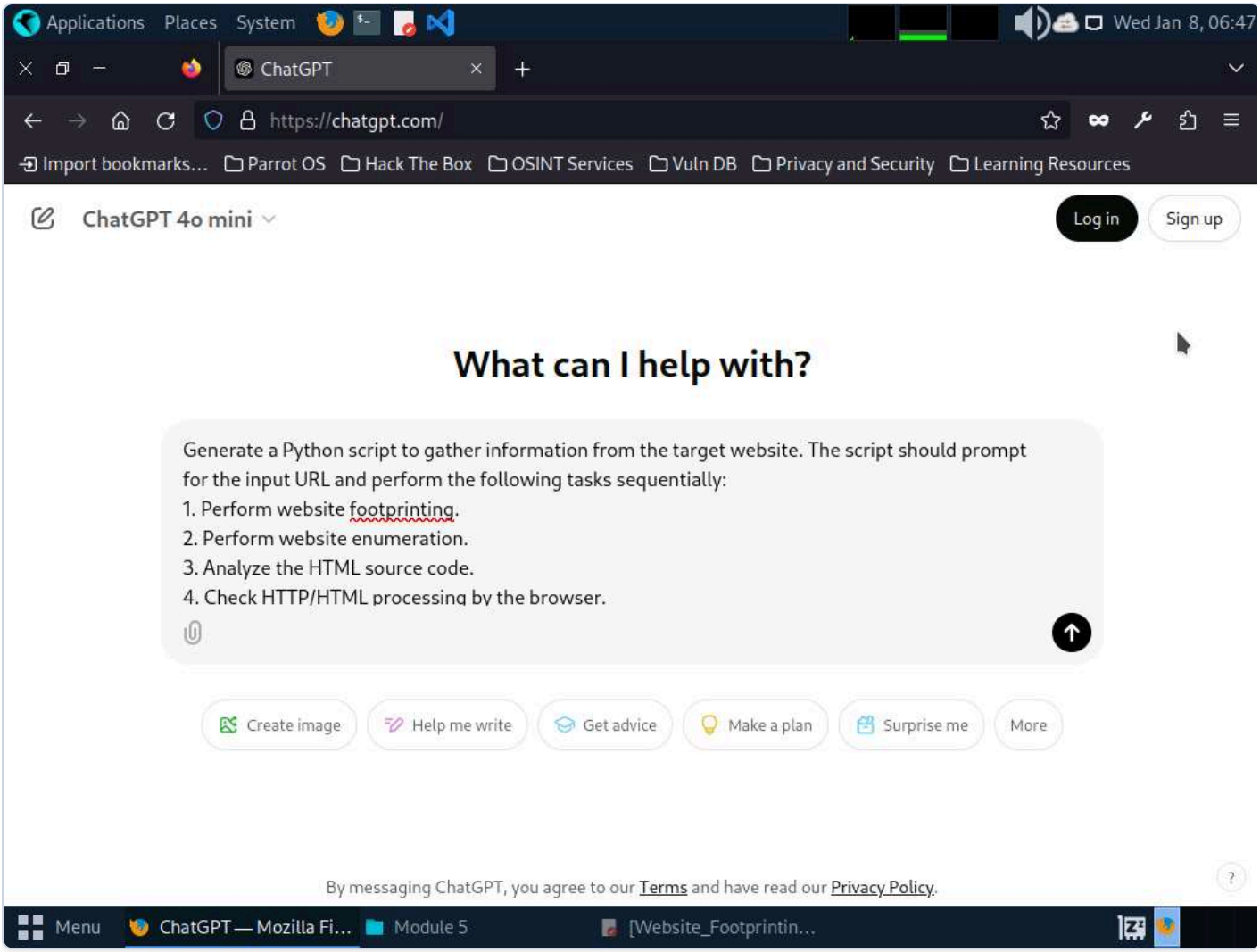
Lab Tasks

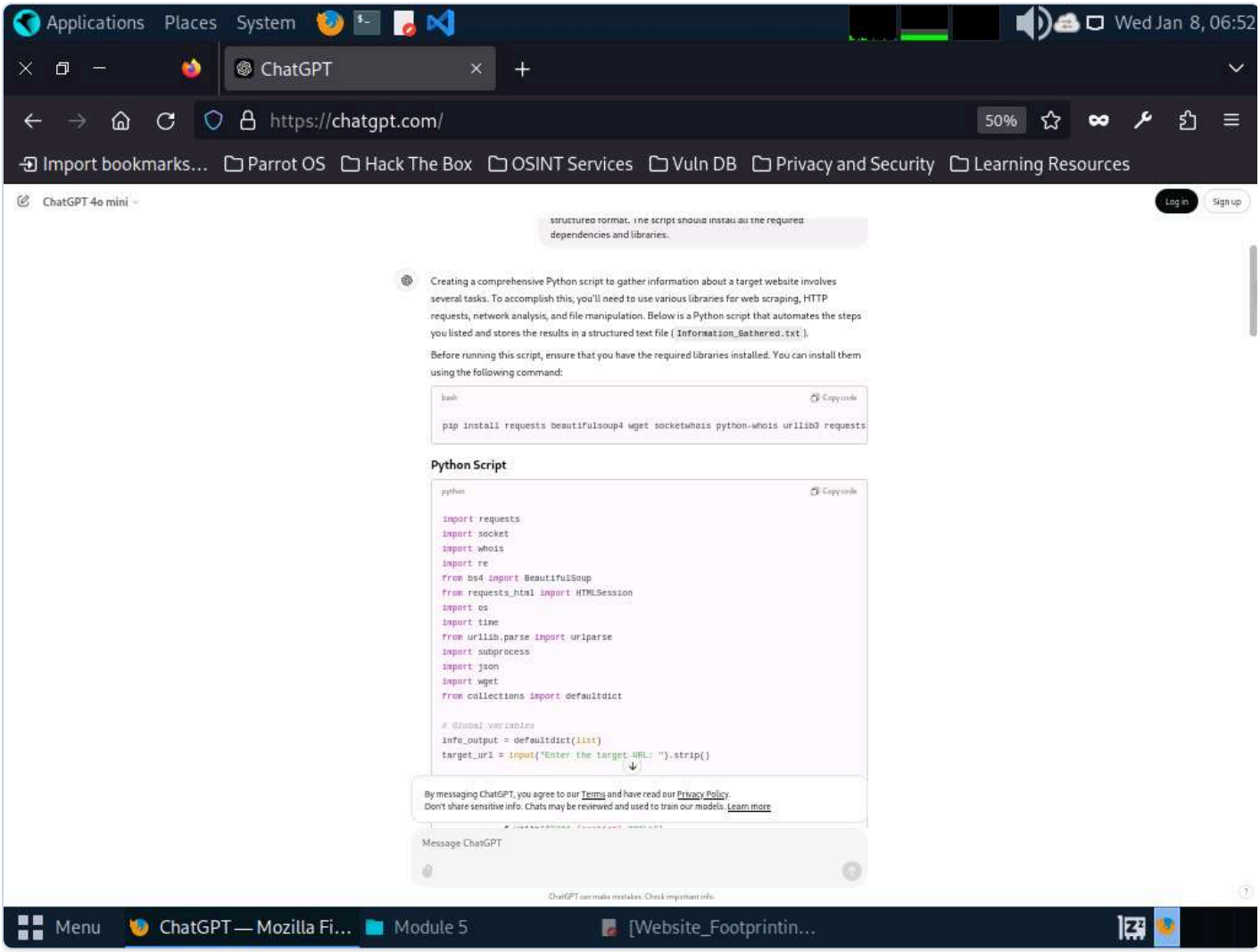
1. In the **Parrot Security** machine, open **Mozilla Firefox**, type **https://chatgpt.com/**, and press Enter.



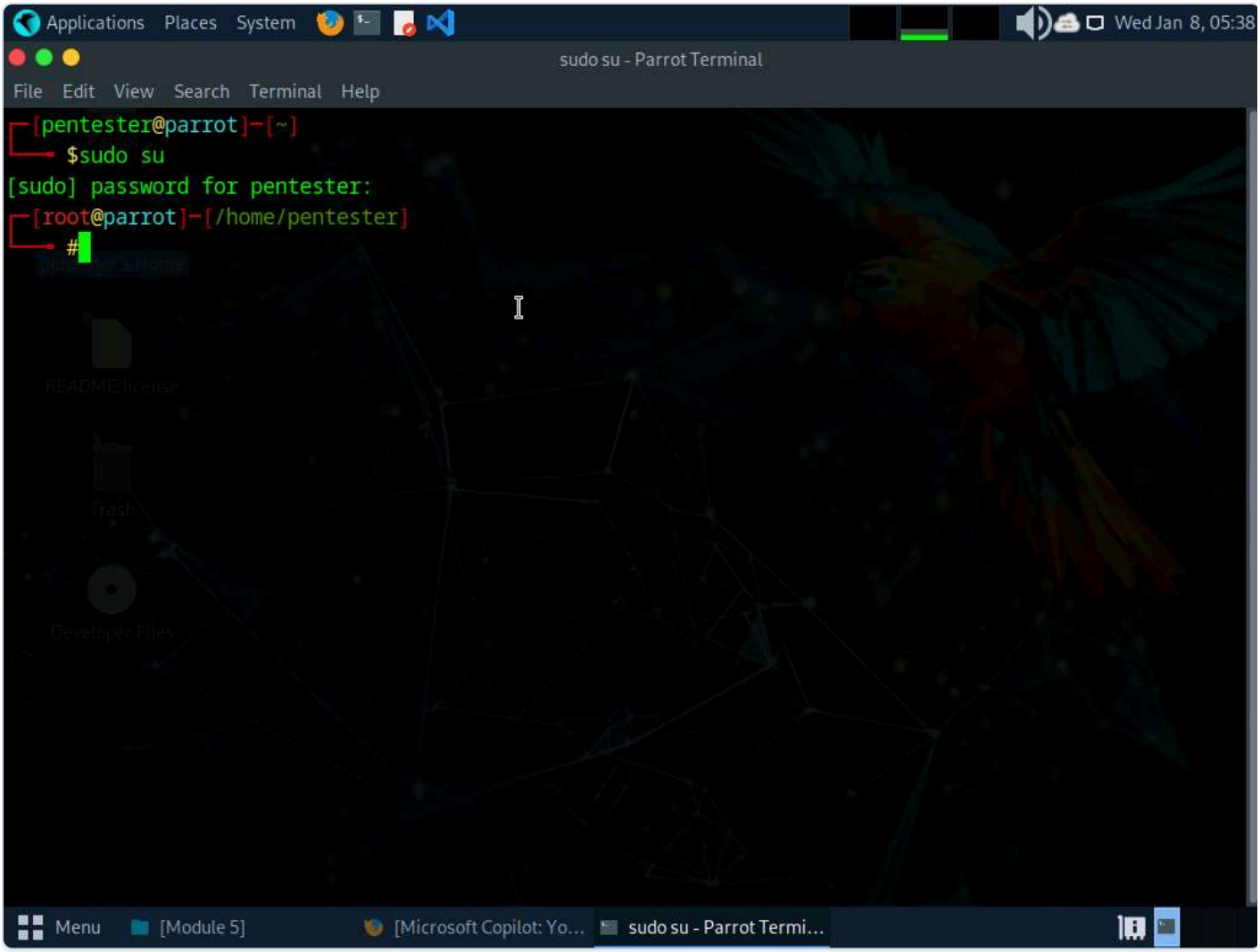
1. In the Message ChatGPT text box enter the below sample prompt and press Enter or click on the Up arrow button for generating a Python script to perform Website Footprinting of a target website:

```
Generate a Python script to gather information from the target website. The script should prompt fo
1. Perform website footprinting.
2. Perform website enumeration.
3. Analyze the HTML source code.
4. Check HTTP/HTML processing by the browser.
5. Identify server-side technologies.
6. Mirror and crawl the website to identify files, directories, and folders.
7. Identify the sitemap.
8. Extract a common word list.
9. Extract metadata and hidden information.
10. Test whether the Target Website is Protected using a Web Application Firewall (WAF).
11. Test whether the Target Website is Protected using Load Balancers.
12. Perform HTTP service discovery.
13. Perform banner grabbing.
14. Enumerate web server directories.
15. Test for proxy functionality.
This script should save the output in 'Information_Gathered.txt' in a structured format. The script
```

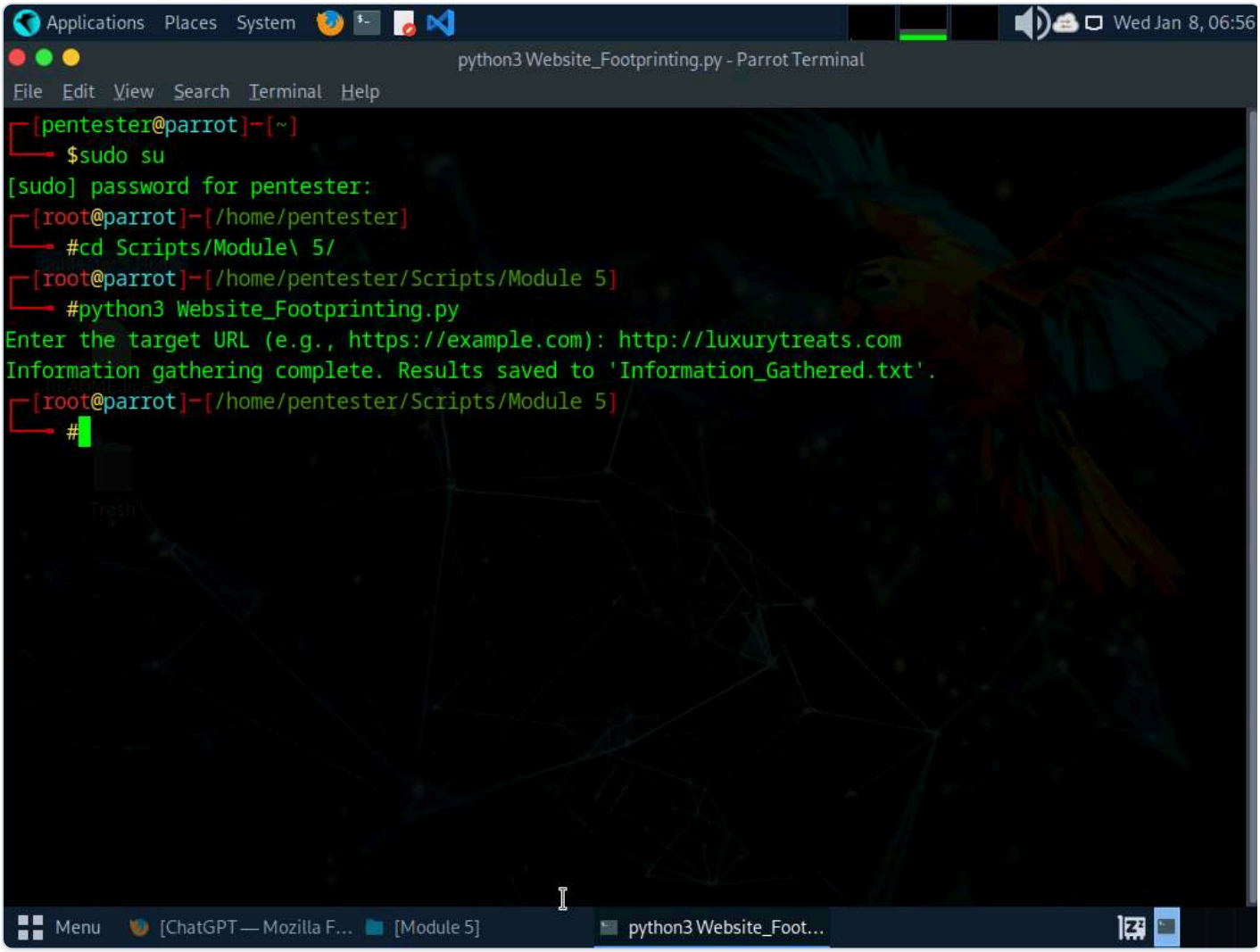

Applications Places System
Wed Jan 8, 06:47
ChatGPT
https://chatgpt.com/
Import bookmarks... Parrot OS Hack The Box OSINT Services Vuln DB Privacy and Security Learning Resources
ChatGPT 4o mini
Log in
Sign up
What can I help with?
Generate a Python script to gather information from the target website. The script should prompt for the input URL and perform the following tasks sequentially:
1. Perform website footprinting.
2. Perform website enumeration.
3. Analyze the HTML source code.
4. Check HTTP/HTML processing by the browser.
Create image
Help me write
Get advice
Make a plan
Surprise me
More
By messaging ChatGPT, you agree to our Terms and have read our Privacy Policy.
Menu
ChatGPT — Mozilla Fi...
Module 5
[Website_Footprintin...



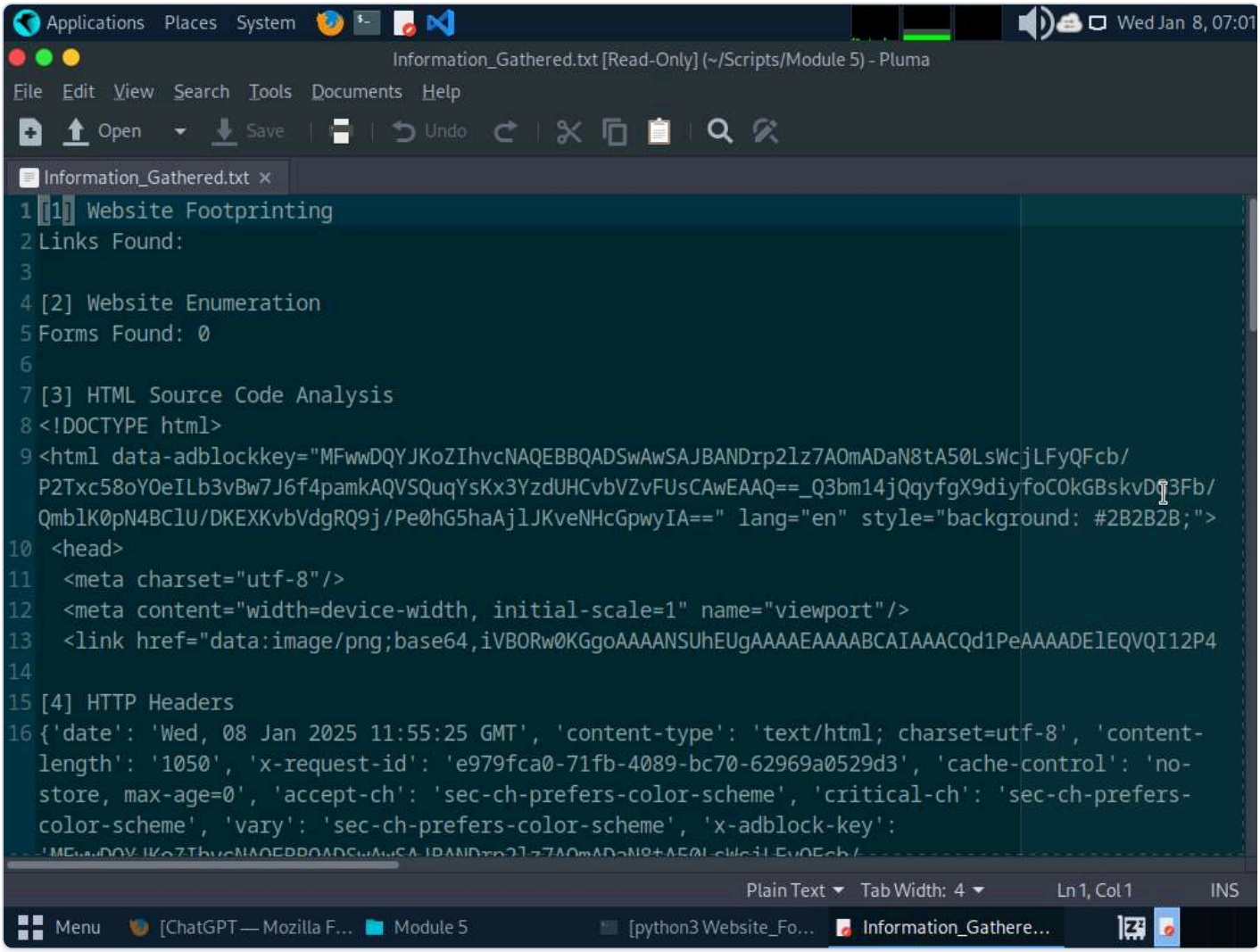
- *(You can get different outputs when performing this lab.)*
- Now, copy and save the generated Python Script in a text file named **Website_Footprinting.py**.
- You can run the sample Python script that is already saved and available in your local repository at: `CPENTv2 Pen Testing Resources/CPENTv2 Scripts/Module 05 Web Application Penetration Testing/Website_Footprinting.py`
- Launch the command-line terminal, type **sudo su** and press Enter. Enter **toor** as the password.

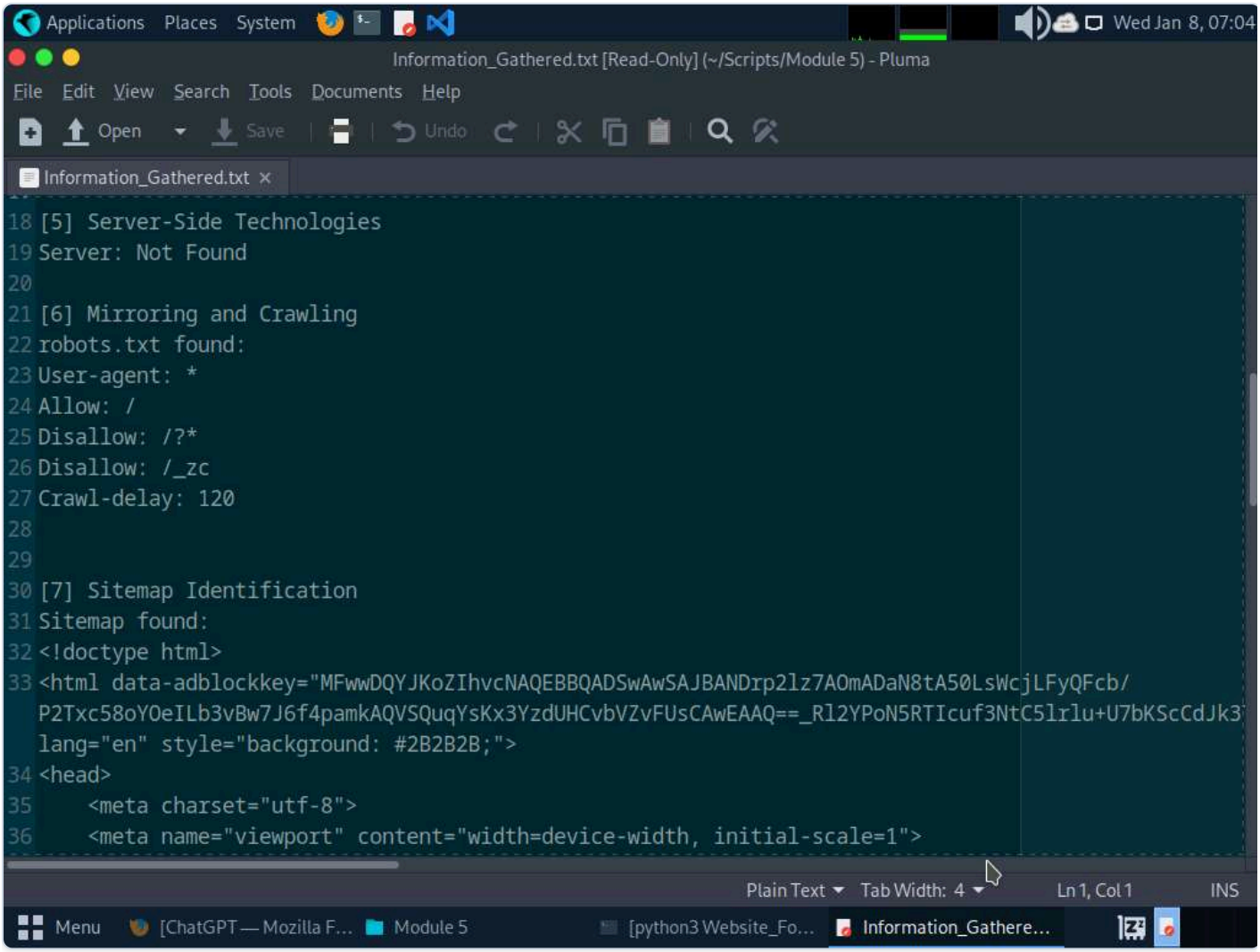


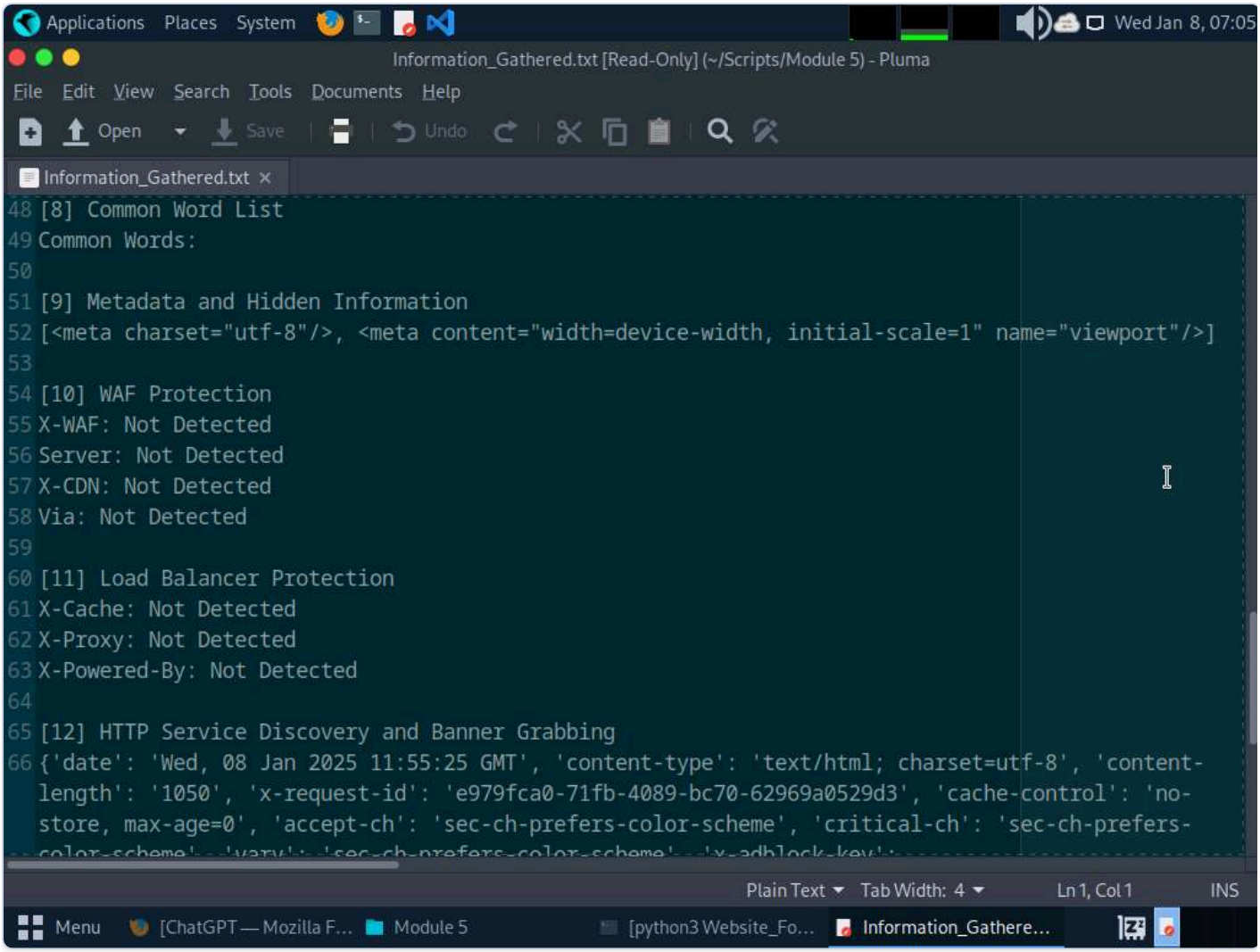
1. Go to the saved script location in the terminal, type **python3 Website_Footprinting.py**, and press Enter. Once executed, it will ask you to enter the target URL.
2. Enter **http://luxurytreats.com** as the target URL.

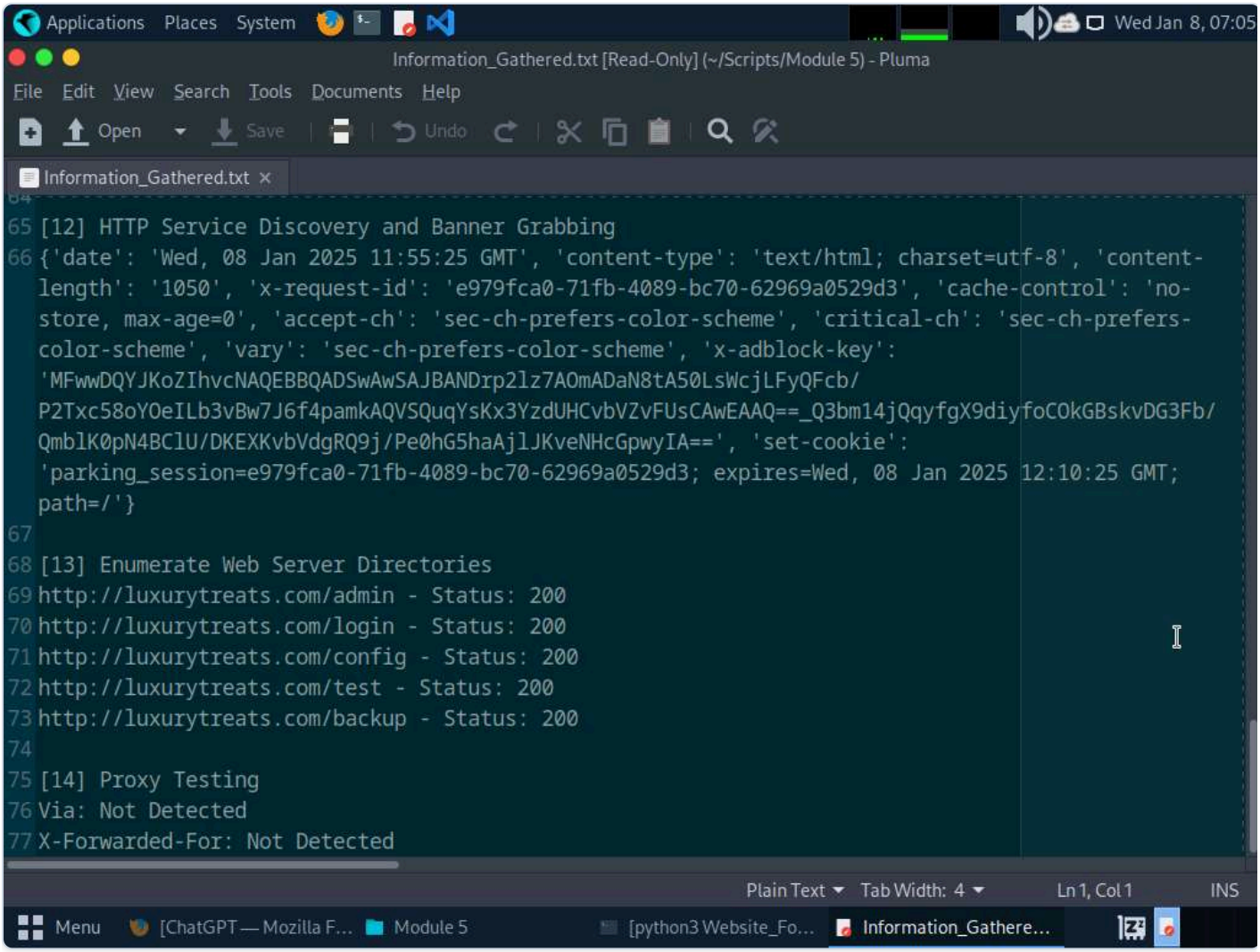


1. After completion of the information gathering, open the **Information_Gathered.txt** file to check for the obtained results.


Applications Places System
Wed Jan 8, 07:01
Information_Gathered.txt [Read-Only] (~/Scripts/Module 5) - Pluma
File Edit View Search Tools Documents Help
Open Save Undo
Information_Gathered.txt
[1] Website Footprinting
Links Found:
[2] Website Enumeration
Forms Found: 0
[3] HTML Source Code Analysis
<!DOCTYPE html>
<html data-adblockkey="MFwwDQYJKoZIhvcNAQEBBQADSwAwSAJBANDrp2lz7AOmADaN8tA50LsWcjLFyQFcb/
P2Txc58oYOeILb3vBw7J6f4pamkAQVSQuqYsKx3YzdUHCvbVZvFUsCAwEAAQ==_Q3bm14jQqyfgX9diyfoCOkGBskvD3Fb/
QmblK0pN4BClU/DKEXKvbVdgRQ9j/Pe0hG5haAj1JKveNHcGpwyIA==" lang="en" style="background: #2B2B2B;">
<head>
<meta charset="utf-8"/>
<meta content="width=device-width, initial-scale=1" name="viewport"/>
<link href="
[4] HTTP Headers
{'date': 'Wed, 08 Jan 2025 11:55:25 GMT', 'content-type': 'text/html; charset=utf-8', 'content-
length': '1050', 'x-request-id': 'e979fca0-71fb-4089-bc70-62969a0529d3', 'cache-control': 'no-
store, max-age=0', 'accept-ch': 'sec-ch-prefers-color-scheme', 'critical-ch': 'sec-ch-prefers-
color-scheme', 'vary': 'sec-ch-prefers-color-scheme', 'x-adblock-key':
Plain Text Tab Width: 4 Ln 1, Col 1 INS
Menu [ChatGPT — Mozilla F... Module 5 [python3 Website_Fo... Information_Gathere...


Applications Places System
Wed Jan 8, 07:04
Information_Gathered.txt [Read-Only] (~/Scripts/Module 5) - Pluma
File Edit View Search Tools Documents Help
Open Save Undo
Information_Gathered.txt
[5] Server-Side Technologies
Server: Not Found
[6] Mirroring and Crawling
robots.txt found:
User-agent: *
Allow: /
Disallow: /?*
Disallow: /_zc
Crawl-delay: 120
[7] Sitemap Identification
Sitemap found:
<!doctype html>
<html data-adblockkey="MFwwDQYJKoZIhvcNAQEBBQADSwAwSAJBANDrp2lz7AOmADaN8tA50LsWcjLFyQFcb/
P2Txc58oYOeILb3vBw7J6f4pamkAQVSQuqYsKx3YzdUHCvbVZvFUsCAwEAAQ==_Rl2YPoN5RTIcuf3NtC5lrlu+U7bKScCdJk3
lang="en" style="background: #2B2B2B;">
<head>
<meta charset="utf-8">
<meta name="viewport" content="width=device-width, initial-scale=1">
Plain Text Tab Width: 4 Ln 1, Col 1 INS
Menu [ChatGPT — Mozilla F... Module 5 [python3 Website_Fo... Information_Gathere...


Applications Places System
Wed Jan 8, 07:05
Information_Gathered.txt [Read-Only] (~/Scripts/Module 5) - Pluma
File Edit View Search Tools Documents Help
Open Save Undo
Information_Gathered.txt
48 [8] Common Word List
49 Common Words:
50
51 [9] Metadata and Hidden Information
52 [<meta charset="utf-8"/>, <meta content="width=device-width, initial-scale=1" name="viewport"/>]
53
54 [10] WAF Protection
55 X-WAF: Not Detected
56 Server: Not Detected
57 X-CDN: Not Detected
58 Via: Not Detected
59
60 [11] Load Balancer Protection
61 X-Cache: Not Detected
62 X-Proxy: Not Detected
63 X-Powered-By: Not Detected
64
65 [12] HTTP Service Discovery and Banner Grabbing
66 {'date': 'Wed, 08 Jan 2025 11:55:25 GMT', 'content-type': 'text/html; charset=utf-8', 'content-length': '1050', 'x-request-id': 'e979fca0-71fb-4089-bc70-62969a0529d3', 'cache-control': 'no-store, max-age=0', 'accept-ch': 'sec-ch-prefers-color-scheme', 'critical-ch': 'sec-ch-prefers-
Plain Text Tab Width: 4 Ln 1, Col 1 INS
Menu [ChatGPT — Mozilla F... Module 5 [python3 Website_Fo... Information_Gathere...



This concludes the lab exercise. Close all open windows.

> **IMPORTANT:** ❓ **Question 5.2.1.1** On the Parrot Security machine, utilize AI tools such as ChatGPT to generate a Python script for conducting comprehensive website footprinting and enumeration on `www.luxurytreats.com`. Execute the `Website_Footprinting.py` script against `www.luxurytreats.com`, and document the number of enumerated web server directories. [!SUCCESS] ✅ **Answer:** **5**